F172 Unit Masterclass | Genetic Engineering & Gene Therapy

OCR AAQ Level 3 Human Biology (Unit F172: Genetics) • Lessons 9 to 12

Biomedical Mechanisms

Lesson 9: Viral & Non-Viral Vector Delivery Systems

Academic Frameworks

Lesson 10: Clinical Trials & AMT-130 Huntington's Therapy

Navy Badges

Lesson 11: Genetic Engineering vs. Gene Therapy

Gene Editing

Lesson 12: CRISPR-Cas9 Gene Editing

Therapeutic Successes & Passing Criteria

NEA Task 2 Framework: Direct preparation for Pass, Merit, and Distinction criteria mapping to patient care, genetic counselling, and therapy relevance.

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Lesson 9 | Viral Vectors in Gene Therapy



Adeno-Associated Virus (AAV)

Biomedical Mechanisms

- **High Neuronal Tropism:** Preferentially transduces post-mitotic CNS neurons and retinal cells.
- **Episomal Persistence:** Retained without genomic integration, drastically reducing oncogenic insertion risk.
- **Payload Limitation:** Strict 4.7 kb packaging limit constrains gene payload size.



Lentiviral Vectors

Biomedical Mechanisms

- **Stable Integration:** Permanently inserts into host genomic DNA for dividing cell lineages.
- **Larger Capacity:** Accommodates payloads up to 8-10 kb (ideal for ex vivo CAR-T).
- **Insertional Mutagenesis:** Rare risk of activating oncogenes near insertion sites.



Adenoviral Vectors

Biomedical Mechanisms

- **Transient Expression:** Rapid clearance without integration; ideally suited for vaccines.
- **High Immunogenicity:** Triggers robust innate immune inflammatory responses.

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Lesson 9 | Non-Viral Vectors & Cellular Delivery Mechanisms

Lipid Nanoparticles & Chemical Delivery

Biomedical Mechanisms



Therapeutic Successes & Passing Criteria

Pathologies & Limitations

- **Lipid Nanoparticles (LNPs):** Ionizable lipid shell encapsulates negatively charged mRNA/DNA.
- **Endosomal Escape:** Facilitates cytoplasmic payload release via membrane fusion.
- **Low Immunogenicity:** Minimal immune rejection compared to viral protein capsids.
- **BBB Challenge:** Systemic LNPs cannot easily cross the blood-brain barrier.

Physical Delivery Modalities

Biomedical Mechanisms



- **Electroporation:** Pulsed high-voltage electric fields create transient cell membrane pores.
- **CED Infusion:** Pressure-driven continuous microinfusion directly into deep brain structures.
- **Stereotactic Guidance:** Real-time neurosurgical MRI targeting bypasses peripheral immune clearance.

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Lesson 10 | Clinical Trial Architecture & The NEA Task 2 Framework

Phase I - Safety & Tolerability

Small Cohorts (N=10-30): Tests safety profile, dose escalation, and adverse event frequency.

Phase II - Biomarker & Efficacy Proof

Intermediate Cohorts: Evaluates target engagement, optimal dosing, and biomarker response.

Phase III - Multi-Centre RCT

Large Scale (N=100s-1000s): Double-blind randomised placebo control measuring definitive clinical endpoints.

NEA Task 2 Assessment Rubric

Pass (P5): Use research to describe relevance of at least two gene therapies for a genetic disorder.

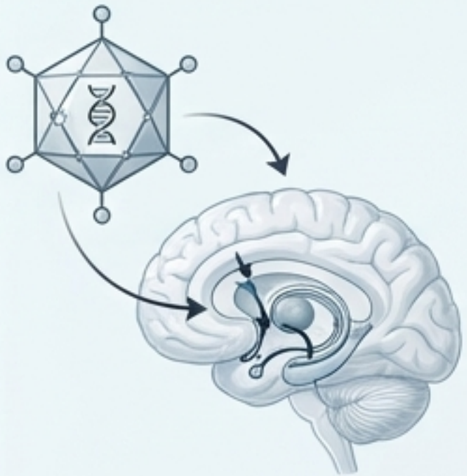
Merit (M3): Extend P5 by deepening the therapeutic context, detailing why therapies are relevant.

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Lesson 10 | AMT-130 Huntington's Clinical Trial Case Study

Therapy Profile

Biomedical Mechanisms



The Target: Autosomal dominant mutation producing toxic huntingtin (mHTT) protein.

The Payload: MicroRNA (miRNA) designed to silence mutant HTT gene expression.

The Vector: AAV5 viral vector chosen for deep neuronal tissue penetration.

NEA Alignment

Therapeutic Successes & Passing Criteria

Pathologies & Limitations

Delivery (P7): Explain method of delivery.

AMT-130 requires MRI-guided neurosurgical infusion (CED) into the striatum.

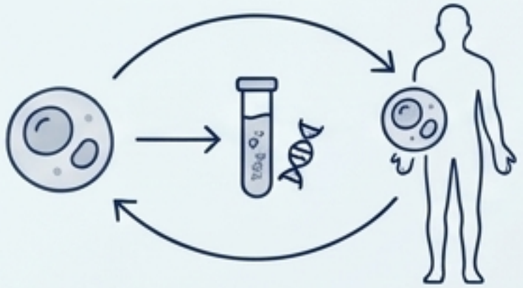
Challenges (M4): Analyse challenges.

Highlights invasive delivery, blood-brain barrier navigation, and irreversible gene silencing risks.

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Lesson 10 | Mapping Gene Delivery Strategies

Ex Vivo (In Vitro)

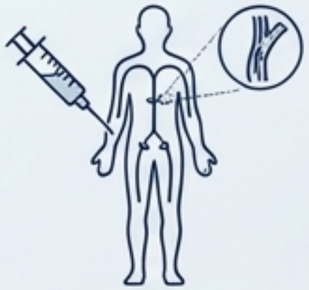


Mechanism: Cells removed, genetically modified outside the body, then reimplanted.

Best For: Hematopoietic stem cells and CAR-T cancer immunotherapies.

Advantage: Highly controlled environment with no systemic immune reaction.

In Vivo



Mechanism: Systemic IV administration of vectors directly into the bloodstream.

Best For: Widespread systemic disorders like Spinal Muscular Atrophy (SMA).

Challenge: High risk of rapid vector clearance by immune system.

In Situ



Mechanism: Localised, direct injection of vectors into specific target tissues.

Best For: Retinal diseases (e.g., Luxturna) or localized tumor treatments.

Challenge: Therapy remains strictly confined to the injection site.

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Lesson 10 | Cellular Targets & Genetic Counselling

Biomedical Mechanisms

Somatic Therapy



Target: Modifies only fully differentiated, non-reproductive patient cells.

Inheritance: Genetic changes cannot be passed to future generations.

Pathologies & Limitations

Germline Therapy



Target: Modifies sperm, egg, or early embryonic cells.

Ethics: Highly controversial; changes are permanently inherited by descendants.

Patient Support & NEA Alignment

Therapeutic Successes & Passing Criteria

Support (P8): Use research to summarise how counsellors explain risks, inheritance patterns, and treatment viability.

Benefit (M5): Explain how counselling is beneficial for informed consent and psychological preparedness.

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Lesson 11 | Defining the Boundaries: Engineering vs. Therapy

Genetic Engineering

Definition: Broad manipulation of an organism's genome using recombinant DNA technologies.

Scope: Encompasses agriculture (GMOs), biomanufacturing (insulin), and basic research.

Focus: Creating novel traits or biological capabilities across species barriers.

Gene Therapy

Definition: A highly specific medical subset of genetic engineering applied to humans.

Scope: Strictly focused on treating or curing underlying genetic pathologies.

Focus: Restoring normal physiological function, never enhancing human traits.

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Lesson 11 | The Core Differences Matrix

	Genetic Engineering	Gene Therapy
Techniques	Often uses plasmids, bacterial transformation, or bio-ballistics (gene guns).	Primarily relies on highly evolved viral vectors or targeted lipid nanoparticles.
Purpose	Maximize crop yield, create biological models, or synthesize industrial proteins.	Silence toxic genes, replace missing genes, or reprogram immune responses.
Ethics	Debates center on ecological impact, corporate monopolies, and biodiversity.	Debates center on patient safety, equity of medical access, and enhancement fears.

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Lesson 12 | CRISPR-Cas9: The Molecular Scalpel

Step 1: The Guide (gRNA)

Targeting: Synthetic RNA molecule engineered to match the target DNA sequence.

Navigation: Leads the Cas9 endonuclease exactly to the desired genomic locus.



Step 2: The Cut (Cas9)

Recognition: Cas9 unwinds DNA and checks for the essential PAM sequence.

Cleavage: Induces a precise double-stranded break at the exact target site.

Step 3: The **Repair** (Cellular Mechanisms)

NHEJ: Non-homologous end joining intentionally disrupts or silences a faulty gene.

HDR: Homology-directed repair inserts a corrected, functional replacement sequence.

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Lesson 12 | Evaluating CRISPR in a Clinical Context

Clinical Potential

- **Permanent Cure**: Can definitively correct mutations at the source (e.g., Sickle Cell Disease).
- **Multiplexing**: Uniquely capable of editing multiple genetic targets simultaneously.

Therapeutic Challenges

- **Off-Target Effects**: Risk of unintended catastrophic cuts in healthy parts of the genome.
- **Delivery Hurdles**: Cas9 proteins are large, making viral vector packaging highly difficult.

Distinction Assessment Framework

Relevance (D3): Discuss if an advanced therapy like CRISPR is the most appropriate option.

Evaluation (D4/D5): Assess three different options focusing on patient care, well-being, treatments, and cures.

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Lessons 9-12 | NEA Task 2 Master Synthesis

Step 1: Pass Criteria (P5, P7, P8)

Action: Use research to describe relevance of two gene therapies.

Detail: Explain vector delivery methods and summarise how genetic counselling assists the patient.

Step 2: Merit Criteria (M3, M4, M5)

Action: Deepen therapeutic context (extend P5).

Detail: Analyse severe vector/delivery challenges and explain the specific benefits of counselling.

Step 3: Distinction Criteria (D3, D4, D5)

Action: Critically discuss therapy relevance against alternative treatments.

Detail: Assess three distinct holistic options covering patient care, well-being, and ultimate cures.

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